

Governed Execution Framework (GEF)

Deterministic Supervision

and

Execution Governance in Probabilistic AI Systems

- **Executive Summary:** Highlighting the shift from additive prompt-wrapper guardrails to structural, deterministic supervision.
- **The Ambient Authority Problem:** Explaining why probabilistic LLMs cannot hold execution keys or ambient capabilities.
- **Architectural Decoupling:** Defining the strict boundary between reasoning proposals and execution authorization.
- **Deterministic Supervision Boundaries:** Explaining the supervisor gate, hard operational limits, and zero ambient authority.
- **Auditable Traceability:** Explaining forensic logging and legal defensibility through immutable execution artifacts.